

NLP for Teens

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■ • What are tokens?

Tokens are the small pieces of text that an LLM actually reads.
LLMs don't understand full words — they understand tokens.

■ · Why LLMs need tokenization

LLMs cannot read full words the way humans do — they only understand tokens, which are small pieces of text.

Tokenization is the step that converts raw text into these tokens so the model can process language correctly.

• WordPiece (BERT)

WordPiece is the tokenizer used in BERT.

Its job is simple: break words into meaningful subword pieces so the model can understand both common and rare words.

■ • BPE (GPT)

BPE is a simple way for GPT to break words into smaller pieces.

Think of it like this:

- Start with letters

- Find pairs of letters that appear a lot

- Join those pairs together

- Keep joining until you get useful pieces (subwords)

■ · BPE (GPT)

Examples:

“international” → inter, national

“lowercase” → lower, case

“transformers” → trans, former, s

■ • BPE (GPT)

BPE looks at all the text it was trained on (books, websites, articles, etc.) and checks:

Which pairs of characters appear the most?

If a pair appears a lot, BPE merges it into a token.

So “useful pieces” = the most frequent patterns in the training data, not necessarily full words.

BPE merges the most frequent character pairs in its training data.

That’s why it creates pieces like “trans”, “former”, “s”.

They appear a lot, so they become tokens — even if they are not full words.

If you want, I can show a tiny fake dataset and walk through BPE step-by-step.

■ Transformers

Transformers are neural networks that understand relationships between words using attention, not recurrence.

They read all words at the same time, not one-by-one like RNN/LSTM.

Transformers introduced self-attention, which lets every word look at every other word.

■ Transformer model types

1. Encoder-only → understanding

BERT, RoBERTa, DistilBERT

Used for: classification, NER, sentiment, embeddings

2. Decoder-only → text generation

GPT, LLaMA

Used for: chatbots, writing, code

3. Encoder-Decoder → translation, summarization

T5, BART

Used for: seq-to-seq tasks

■ BART

Bidirectional
Auto-
Regressive
Transformer

Bidirectional → like BERT (reads the whole sentence at once)

Autoregressive → like GPT (generates text one token at a time)

So BART = BERT + GPT in one model.

■ Quick comparison table

Tokenizer	Used in	How it works
WordPiece	BERT	Splits words into subwords using "##"
BPE	GPT	Merges most-frequent character pairs
SentencePiece	T5	Learns tokens from raw text, uses " _ "

Assignment

- Use the sentence:
 - The model learns patterns from large amounts of text.
- Load BERT (WordPiece), GPT-2 (BPE), BART (BPE), T5 (SentencePiece).
- Tokenize the sentence with all four.
- Encode → IDs, decode → text.
- Compare the result

Summary

- ✓ Tokens small text pieces LLMs read
- ✓ WordPiece: splits words into subwords using “##”
- ✓ BPE merges most-frequent character pairs
- ✓ SentencePiece: learns tokens from raw text using “_” for spaces

Thank You

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